Pivotal-Based Classical and Bayesian Estimation for the Lomax Distribution under Block Generalized Progressive Hybrid Censoring

DA 2141 Statistics for Data Science · Course project

[Student name(s) and roll number(s)] · October 2026

MLE | Pivotal | Bayes (joint prior) | Order-restricted

The project at a glance

1

Background & likelihood

Lomax model, block GPHC scheme, likelihood function

2

Estimation

MLE, pivotal, Bayes (joint prior), order-restricted MLE & Bayes

3

Simulation

18 designs × 2,000 replications, 12 estimators

4

Real data

British coal-mining disasters in 3 periods

5

Report & code

Reproducible Python package + 29-page report

New for Lomax: an MLE existence condition, a decreasing & bounded pivot (truncated algorithm), and an exact importance sampler for joint and ordered priors.

Why this problem?

1 Several test facilities

- Not enough equipment to test all n units at once → split into k blocks (block censoring)
- Identical-looking facilities still give different results → model a facility effect α_i
- Common failure mechanism → shared parameter λ

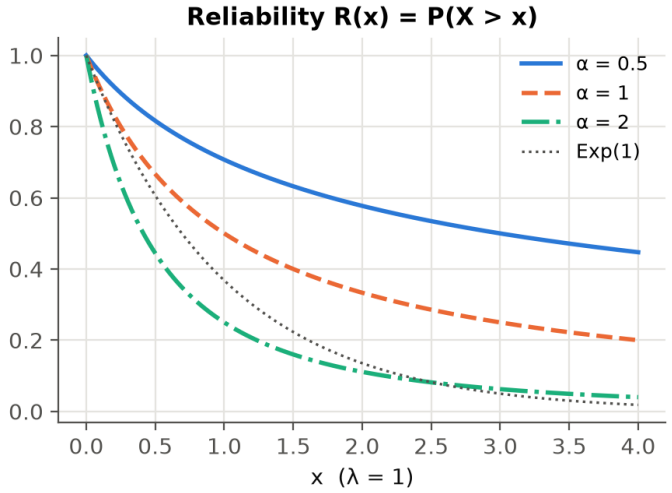
2 Stop the test early (censoring)

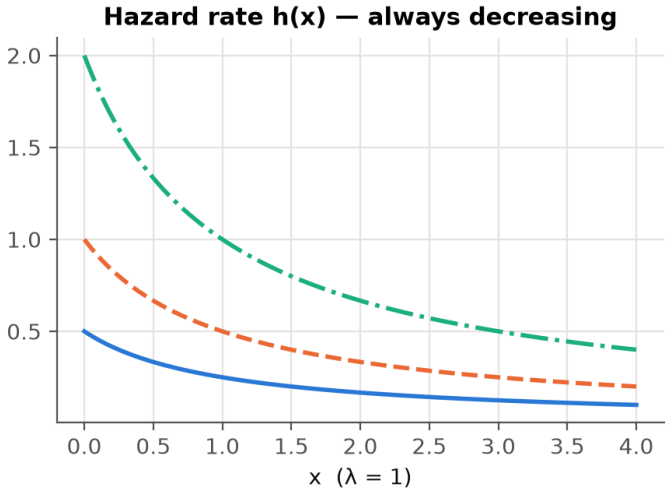
- Type-I (fixed time) may give too few failures; Type-II (fixed count) may take too long
- Generalized progressive hybrid censoring: at least κ failures, stop by time T or the m-th failure
- Units may be withdrawn during the test (progressive removals)

Goal: estimate the facility effects α_i, the common λ, and the reliability R(t), hazard h(t) and median lifetime of each facility — with point and interval estimates, with and without the natural ordering $\alpha_1 \le \alpha_2 \le \alpha_3$.

The Lomax (Pareto II) lifetime model

$$f(x) = \alpha\lambda(1+\lambda x)^{-(\alpha+1)}, \quad R(x) = (1+\lambda x)^{-\alpha}, \quad h(x) = \frac{\alpha\lambda}{1+\lambda x}$$

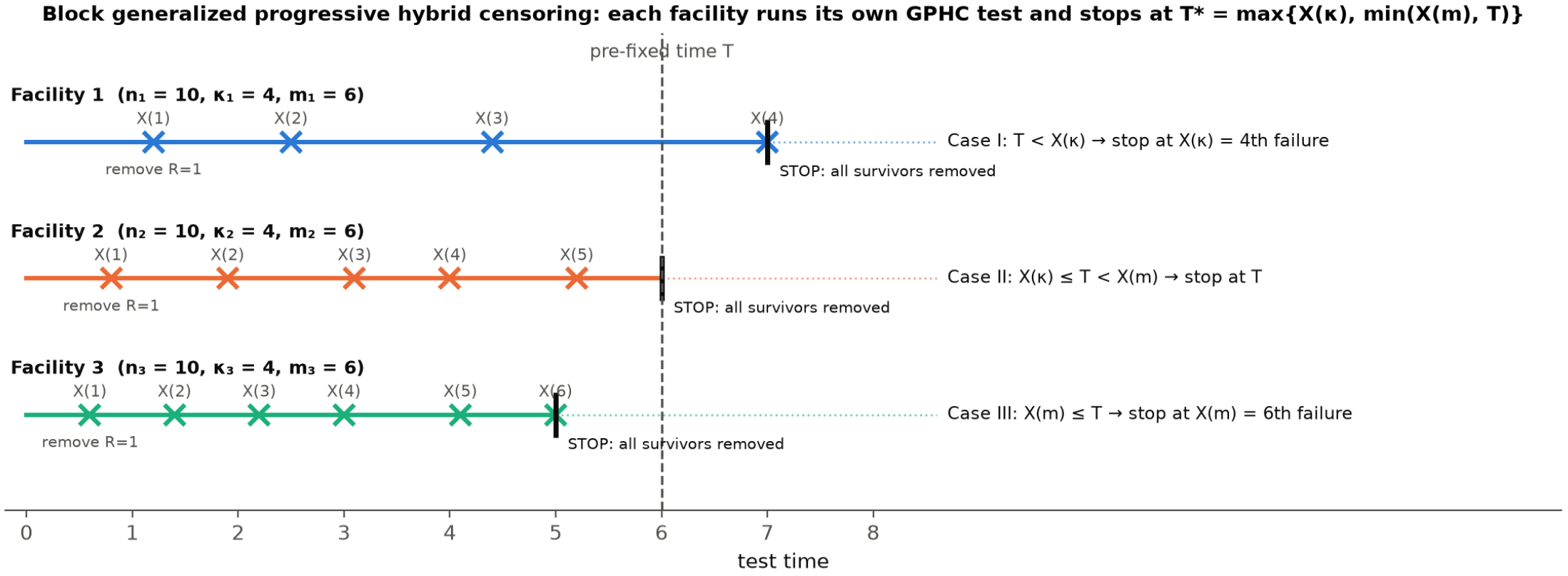




- Heavy (polynomial) right tail; mean infinite if α ≤ 1
- Hazard always decreasing
- Gamma mixture of exponentials; exceedances over a high threshold
- Exponential limit: λ → 0, α → ∞
- Key trick:

$$U = \alpha \log(1+\lambda X) \sim \text{Exp}(1)$$

Block generalized progressive hybrid censoring (BGPHC)



$$T_i^* = \max\left\{X_{\kappa_i : m_i : n_i},\ \min\left(X_{m_i : m_i : n_i},\ T_i\right)\right\}$$

Case I: stop at the κ-th failure · Case II: stop at T · Case III: stop at the m-th failure. Each facility i records D_i failures, removals R_{ij} and R_i^T units withdrawn at T.

One likelihood covers all three cases

$$L(\alpha, \lambda) \;\propto\; \prod_{i=1}^{k} \alpha_i^{D_i}\; \lambda^{D} \prod_{i,j} (1+\lambda x_{ij})^{-1}\; \exp\left\{-\sum_i \alpha_i \Delta_i(\lambda)\right\}$$

$$\Delta_i(\lambda) = \sum_{j=1}^{D_i} (1+R_{ij}) \log(1+\lambda x_{ij}) + R_i^{T} \log(1+\lambda T_i)$$

Key insight

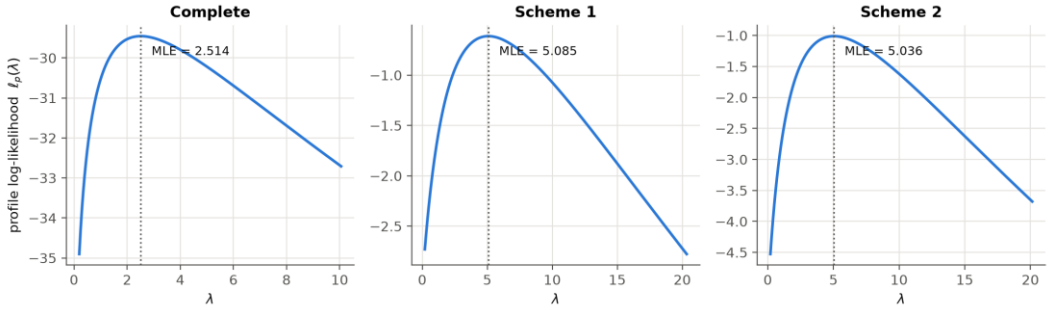
- $\Delta_i(\lambda)$ is the total time on test on the exponential scale $\log(1+\lambda x)$
- For fixed λ, α_i appears only as $\alpha_i^{D_i} e^{-\alpha_i \Delta_i}$ — a gamma kernel
- → closed-form MLE of α_i, chi-square pivots, gamma posterior conditionals, isotonic order-restricted solution

Maximum likelihood estimation

$$\hat{\alpha}_i(\lambda) = \frac{D_i}{\Delta_i(\lambda)}, \qquad \hat{\lambda} = \arg\max_{\lambda} \ \ell_p(\lambda)$$

$$S_0 = \sum_i \frac{D_i B_i}{2A_i} - \sum_{i,j} x_{ij} > 0 \ \Rightarrow \ \hat{\lambda} \in (0, \infty) \text{ exists}$$

- Profile likelihood ℓp(λ) → 1-D search (grid + Brent)
- Theorem 2: MLE exists if $S_0 > 0$ — for one complete sample this is CV > 1 (data more variable than exponential)
- Otherwise λ̂ → 0, α̂ → ∞ (exponential limit)
- Asymptotic CIs: observed Fisher information + delta method for R(t), h(t), MdTF



Profile log-likelihood of λ, coal-mining data (complete, Scheme 1, Scheme 2): a single maximum

Pivotal (generalized pivotal quantity) inference

$$\mathcal{P}(\lambda) = 2\sum_{i}\sum_{j<D_i} \log\frac{W_{iD_i}}{W_{ij}} \sim \chi^2_{2\sum(D_i-1)}, \qquad 2\alpha_i\Delta_i(\lambda) \sim \chi^2_{2D_i}$$

Theorem 3 (new for Lomax)

- $\mathcal{P}(\lambda)$ is strictly decreasing in λ
- $\mathcal{P}(\lambda) \to 0$ as $\lambda \to \infty$, but $\mathcal{P}(0^+)$ is finite (exponential limit)
- So $\mathcal{P}(\lambda) = c$ has a root only if $c < \mathcal{P}(0^+)$ → draw c from a truncated χ^2
- IER/Burr-XII pivots (refs 1–2) increase from 0 to ∞ — no truncation needed there

1. Draw $c \sim \chi^2$ truncated to $(0, \mathcal{P}(0^+))$; solve $\mathcal{P}(\lambda) = c$ (bisection)
2. Draw $\chi^2(2D_i)$ and set $\alpha_i = \chi^2 / (2\Delta_i(\lambda))$
3. Repeat N times → point estimate = median, 95% GCI = 2.5% and 97.5% quantiles

Median, not mean: the pivotal draws are very right-skewed.

Bayesian estimation with a joint Gamma–Dirichlet prior

$$\pi(\alpha) \propto \prod_i \alpha_i^{c_i - 1} e^{-b_0 \alpha_+} \alpha_+^{a_0 - c_+}, \qquad \alpha_+ = \sum_i \alpha_i$$

- Sum α_+ ~ Gamma(a_0, b_0); shares α_i/α_+ ~ Dirichlet(c) (ref. 3)
- Dependent prior: correlation > 0 if $a_0 < c_+$, = 0 if $a_0 = c_+$, < 0 if $a_0 > c_+$
- λ ~ Gamma(a, b) — must be proper: vague priors give an improper posterior

$$\hat{\theta}_{SE} = E(\theta \mid \text{data}), \qquad \hat{\theta}_{LX} = -\frac{1}{h} \log E\left(e^{-h\theta} \mid \text{data}\right)$$

Squared-error and LINEX (h = 1) loss; 95% HPD credible intervals

Importance sampler (exact, no MCMC)

1. Sample λ from its 1-D marginal on a fine grid
2. Sample $\alpha_i \mid \lambda$ ~ Gamma($D_i + c_i$, $b_0 + \Delta_i(\lambda)$)
3. Weight w = $\alpha_+^{(a_0 - c_+)}$ → posterior means, LINEX, HPD

No burn-in, no tuning, fully vectorised; exact when $a_0 = c_+$.

Order-restricted inference: $\alpha_1 \le \alpha_2 \le \ldots \le \alpha_k$

19–42%

of simulated samples had unrestricted MLEs in the wrong order

- Stress levels, operators or time periods often have a known ordering
- Coal data: mine safety improved, so tails got lighter

Order-restricted MLE (isotonic regression)

$$\tilde{\alpha}_u(\lambda) = \left[\min_{s \le u} \max_{t \ge u} \frac{\sum_{r=s}^{t} \Delta_r(\lambda)}{\sum_{r=s}^{t} D_r} \right]^{-1}$$

Computed by pool-adjacent-violators; then maximise over λ.

Order-restricted Bayes (ordered GD prior, ref. 3)

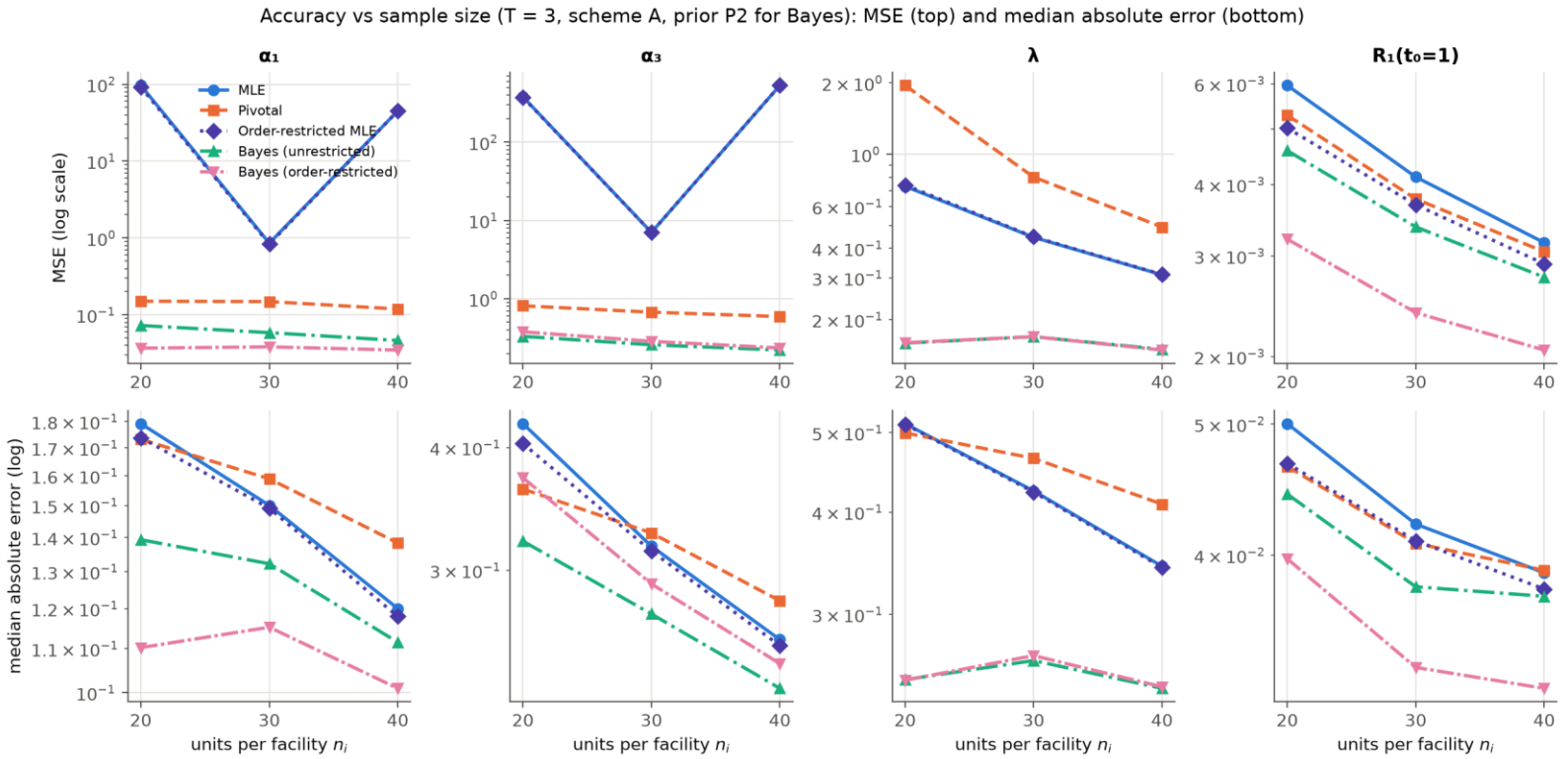
- Prior = density of the order statistics of a GD vector (sum over k! permutations)
- Sampler: draw α_i' from the gamma conditionals, sort them, re-weight by target / density of the sorted vector

Simulation design

Facilities / truth	k = 3; α = (0.50, 0.75, 1.00) (ordered), λ = 1, t_o = 1
Sizes (n_i, m_i, κ_i)	S1 = (20, 15, 10), S2 = (30, 24, 15), S3 = (40, 32, 20)
Time limit	T = 3 (short test), T = 8 (long test)
Removal schemes	A: all at m-th failure; B: all at 1st failure; C: half / half
Replications	2,000 per design × 18 designs; 1,000 pivotal draws; 2,000 IS draws
Priors	P1 informative (a_o = 4), P2 weakly informative (a_o = 1)
Criteria	Bias, MSE, median abs. error; length and coverage of 95% intervals

Heavy tails ($\alpha \leq 1$) keep the Lomax clearly away from its exponential limit; a light-tailed pilot (α = 1–2) gave non-existent MLEs in about 20% of samples.

Accuracy: Bayes-OR is best; the MLE has rare but huge failures



96.6 vs 0.15

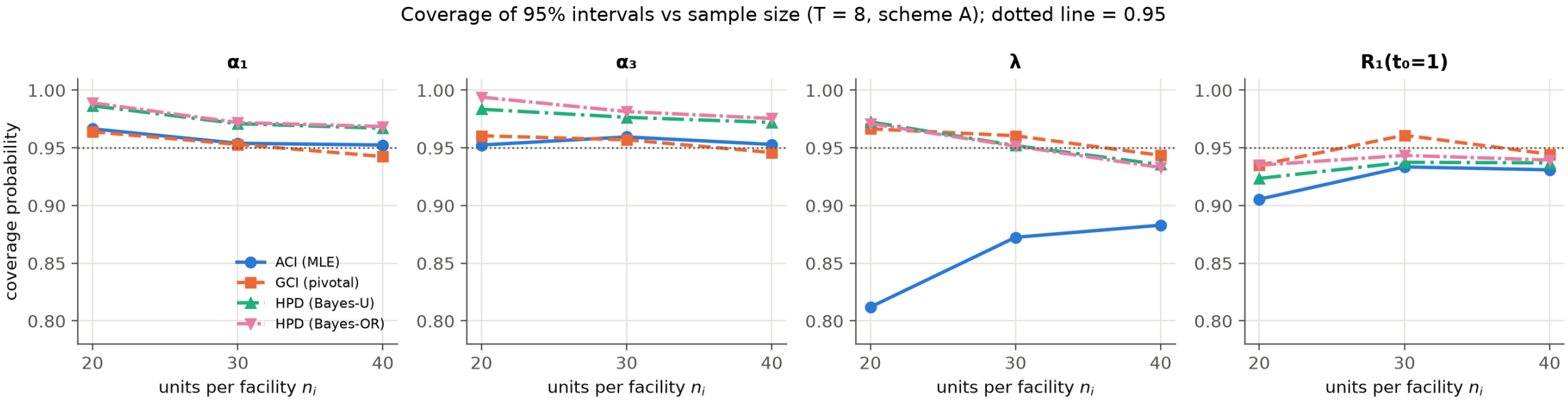
MSE of α_1: MLE vs pivotal (n = 20, T = 3) — same typical error (0.18 vs 0.17)

18 / 18

designs where order-restricted Bayes has the lowest MSE for α_2, R_2, R_3

Bottom row: median absolute error — robust to the few runaway MLEs.

Intervals: pivotal GCIs keep their promise of 95%



93.5–97.5%

coverage of pivotal GCIs in all 18 designs

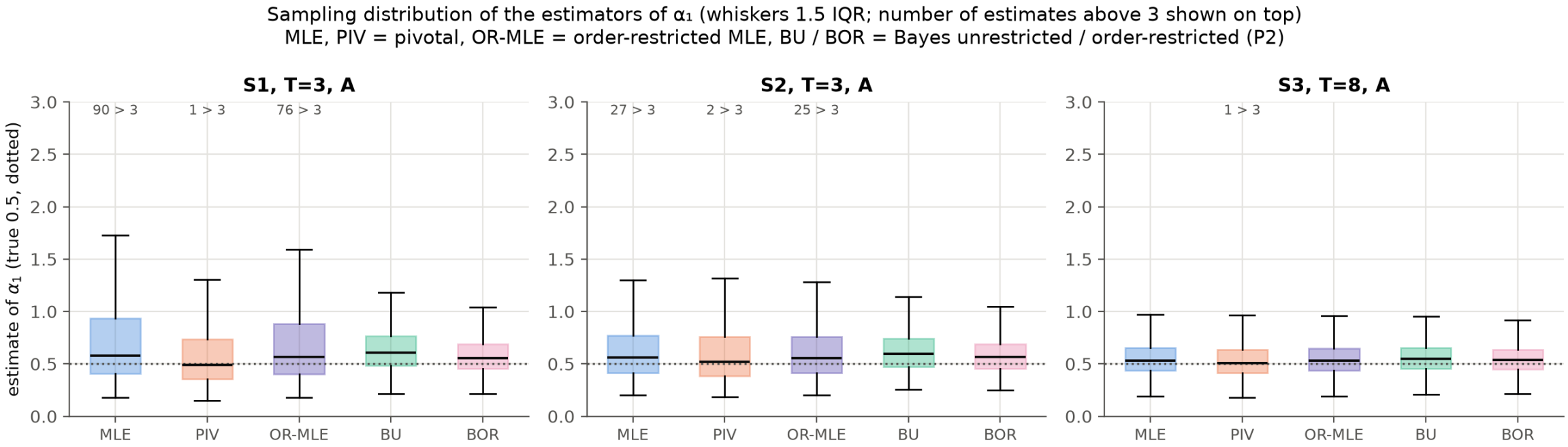
81–90%

coverage of MLE-based ACIs for λ (Wald fails)

Shortest

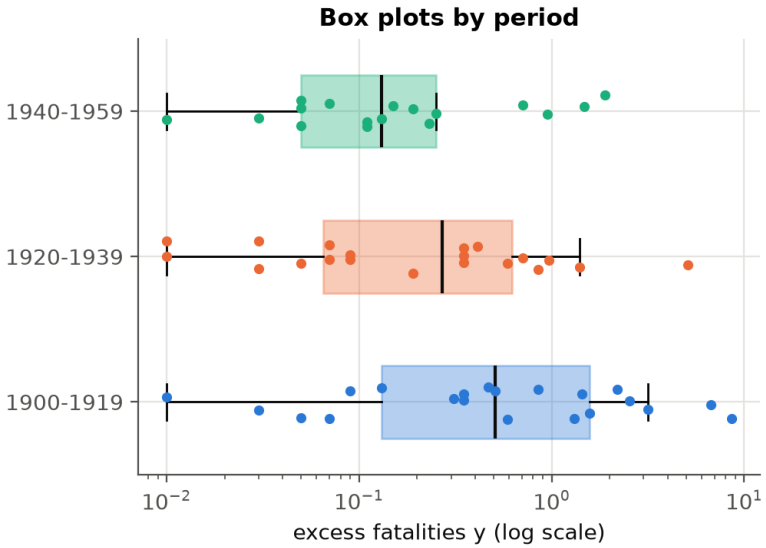
Bayes HPD intervals, coverage ≥ 95% for α, λ; 92–96% for R(t)

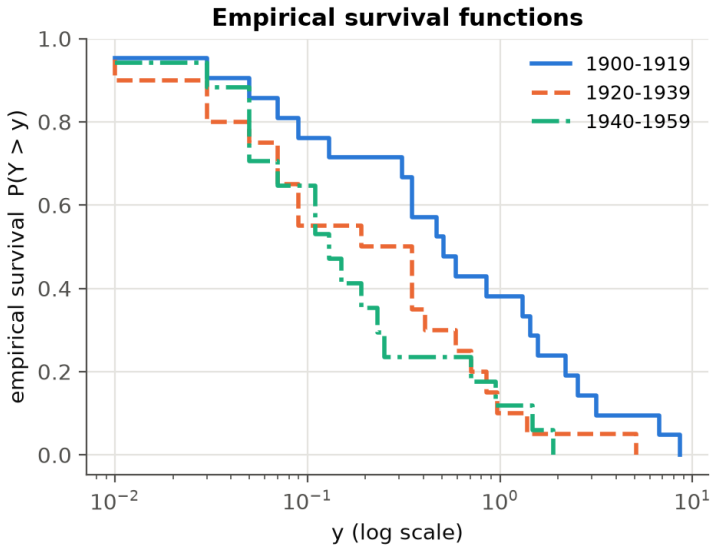
Why the MLE's MSE explodes



- Typical MLE is fine — the medians sit near the truth (0.5)
- But 90 of 2,000 MLEs exceed 3 when $n_i = 20$, $T = 3$: the sample looks exponential, $\hat{\lambda} \to 0$, $\hat{\alpha}$ explodes (Theorem 2)
- Pivotal and Bayes estimators are robust; the problem fades for $n_i \geq 30$ with a long test

Real data: fatalities in British coal-mining disasters

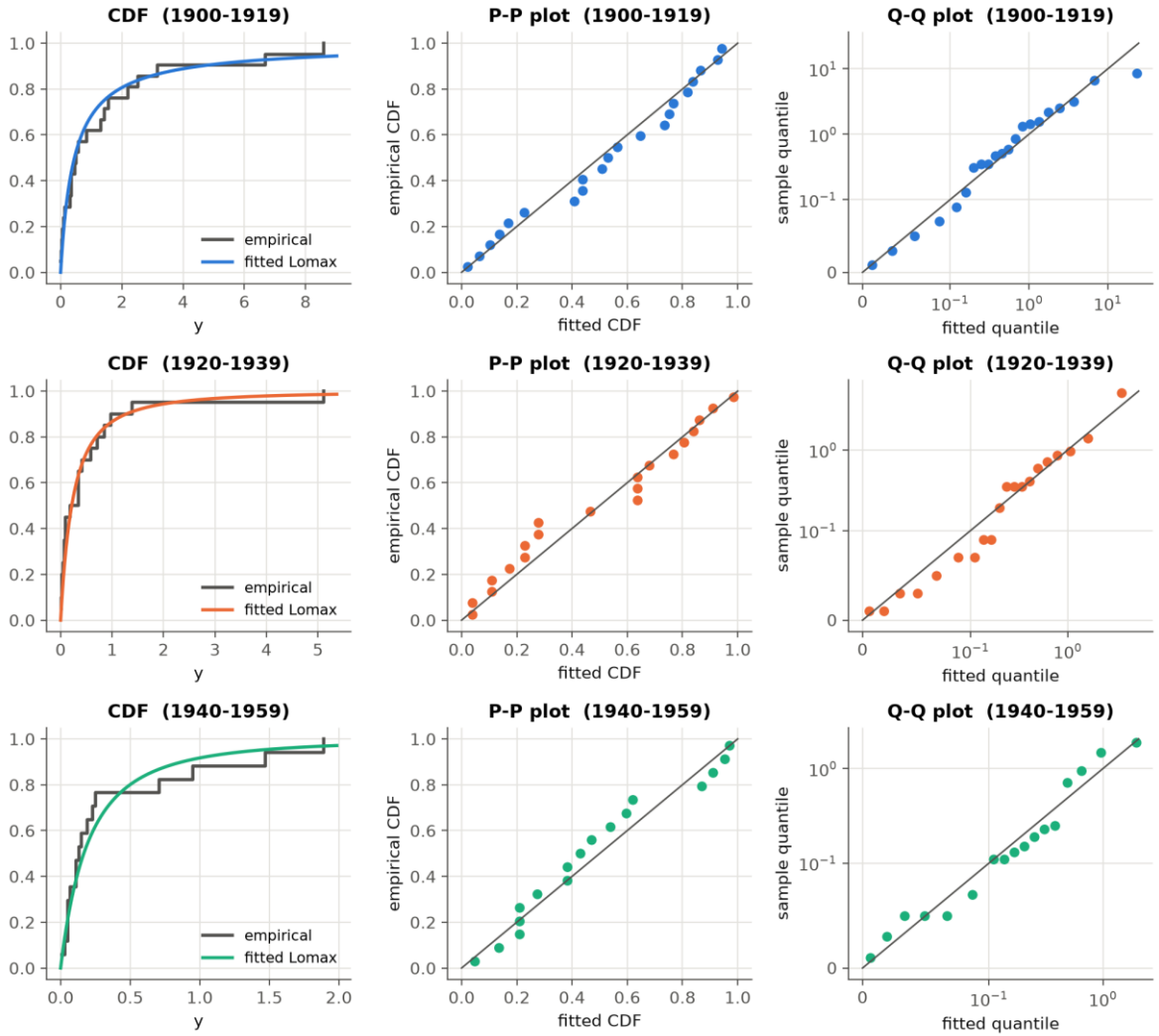




- Disasters with ≥ 10 deaths (data from ref. 2); 3 periods = 3 "facilities": 1900–19 (n = 21), 1920–39 (n = 20), 1940–59 (n = 17)
- Model the excess over the recording threshold: y = x − 0.19 (peaks over threshold)
- Prior knowledge: safety improved, so tails get lighter: $\alpha_1 \le \alpha_2 \le \alpha_3$

All three periods are right-skewed with coefficient of variation > 1 → Lomax MLE exists.

Model checking: Lomax fits, common λ is supported



p > 0.54

KS tests, Lomax with common λ, all periods

p = 0.65

LRT: H_0 $\lambda_1 = \lambda_2 = \lambda_3$ not rejected

p = 0.051

LRT: H_0 $\alpha_1 = \alpha_2 = \alpha_3$ (complete data)

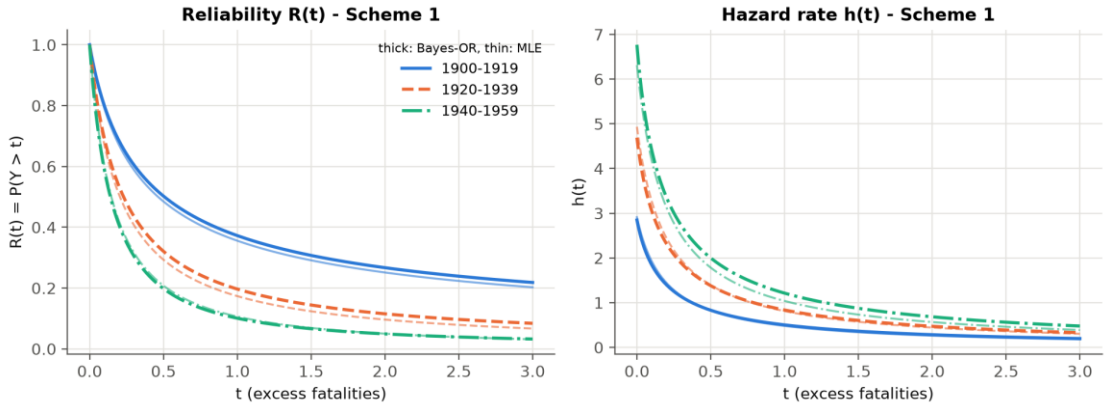
p = 0.026

same test, censored Scheme 2

Lomax beats the exponential on AIC in every period; the exponential is rejected for 1920–39 (KS p = 0.035).

Estimates: large disasters became much less likely

Complete data	MLE	Pivotal	Bayes-U	Bayes-OR
α_1 (1900–19)	0.913	0.794	0.980	0.981
α_2 (1920–39)	1.602	1.342	1.753	1.663
α_3 (1940–59)	1.981	1.638	2.178	2.328
λ	2.514	3.326	2.675	2.663
$R_1(0.5)$	0.476	0.470	0.487	0.488
$R_2(0.5)$	0.271	0.275	0.287	0.303
$R_3(0.5)$	0.199	0.208	0.218	0.192



- P(a disaster kills > ~35 people) fell from ≈ 0.48 (1900–19) to ≈ 0.20 (1940–59)
- Censoring (Schemes 1, 2) widens intervals for λ but barely moves R(t)
- Bayes-OR gives the shortest HPD intervals for the reliabilities

Conclusions and recommendations

Pivotal → honest intervals

Truncated, median-based GPQs give near-95% coverage without priors or large-sample theory.

Order-restricted Bayes → most accurate

The joint GD prior plus a known ordering gives the smallest errors and the shortest intervals.

Be careful with the plain MLE

Lomax shapes need visible tails: small or short tests can make the MLE (and Wald CIs) unreliable.

Future work: exact pivots under GPHC, bootstrap intervals for the OR-MLE, optimal censoring plans, facility-specific λ_i.

References

[1] Kumari, Tripathi, Lodhi & Wang (2025). Pivotal based reliability inference for Burr-XII distribution under block progressive type-II censoring. Comm. Stat. – Simulation & Computation 55(8).

[2] Lodhi, Gangopadhyay & Bhattacharya (2025). Likelihood and pivotal-based reliability inference for inverse exponentiated Rayleigh distribution under block progressive censoring. QTQM 22(4), 779–804.

[3] Sultana, Çetinkaya, Raqab & Kundu (2026). Inference for block Type-I hybrid censored Weibull populations. Comm. Stat. – Theory & Methods 55(11), 3436–3459.

[4] Cho, Sun & Lee (2015). Exact likelihood inference for an exponential parameter under generalized progressive hybrid censoring scheme. Statistical Methodology 23, 18–34. (the one additional reference)

Course texts: Casella & Berger, Statistical Inference; Hogg, McKean & Craig, Introduction to Mathematical Statistics.

Thank you — questions?

Code, results, report and this deck are fully reproducible (fixed random seeds).